

A harmonized panel of United States compression-ignition engine certification data, with resolved certification lineage

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Abstract

The United States Environmental Protection Agency (EPA) publishes emission certification records for compression-ignition engines, but distributes them as two spreadsheets whose internal structures are mutually incompatible and whose true column headers are not on the first row. The result is a public record that is open in principle and costly to use in practice: every researcher must independently reverse-engineer two layouts, and the characteristic failure is not an error message but a quietly wrong number.

We present CIDEX, a harmonized engine-family-level panel reconciling both sources into a single schema: **8,627 engine families**, **9,794 engine configurations** and **94,869 emission records**

from **108 manufacturers**, spanning model years 2011–2027, 14 pollutants and 3 test types.

Beyond reorganisation, we resolve a structure present in the source but not previously published in usable form: **5,982 nonroad families declare a predecessor family**, and these declarations chain. Resolving them yields **2,541 distinct certification lineages** with a maximum depth of **14 model years** and a mean of 1.382, allowing a family’s regulatory history to be followed across more than a decade. We show the lineage graph is a forest, and that 30 families use an undocumented self-reference convention that a naive traversal misreads as cycles.

We further identify 50 records carrying physically impossible negative certification results, and 4 model years whose apparent decline is an artefact of EPA’s archive boundaries rather than a change in the fleet. Both are published as flagged data rather than silently corrected or removed, because a cleaned column conceals a finding and a removed row conceals a defect.

CIDEX preserves certification units as issued and performs no silent conversion: highway results are in g/bhp-hr and nonroad results in g/kW-hr, and we argue that normalising them would convert a visible incompatibility into an invisible one. The paper documents the harmonization decisions, the automated and manual validation, and one defect found during development in which a merge on a non-unique key silently multiplied rows — included because it is the exact class of error this dataset exists to prevent, and because recording how it was caught is more useful to a reader than asserting that the result is correct.

Keywords: engine certification; compression-ignition engines; nonroad engines; emissions; data harmonization; certification lineage; open government data

1 Background and summary

1.1 The record and its unit

No compression-ignition engine may be sold in the United States without a certificate of conformity. The unit of certification is the **engine family** — a group of engines expected to share emission characteristics — rather than an individual engine model. A family is tested at one or more configurations, results are compared against the standard applicable to its service class and model year, and a certificate is issued.

The accumulated record is the only comprehensive public account of what engines were approved to emit, by whom, and under which standard. It bears directly on fleet composition, technology adoption, the realised stringency of successive standards, and the gap between certified and in-service behaviour. It is also, for a large part of the economy, the only fine-grained public description of the equipment itself: displacement, power rating, aftertreatment configuration and test procedure, family by family.

Two properties of the unit shape everything downstream. First, the family is a regulatory construct rather than a physical one, so counts of families are counts of certificates and not of engines or of sales. Second, EPA encodes the model year in the first character of the family name, which makes the name unique within the published window and therefore usable as a key without inventing a surrogate. Both properties are exploited in the schema and are stated here so a reader knows what a row of CIDEX is.

1.2 Open in principle, costly in practice

EPA publishes this record openly, without authentication, refreshed quarterly. The difficulty is structural, and it has three parts.

First, **both workbooks place their column headers on the second row**, reserving the first for a merged group header. The default behaviour of standard spreadsheet readers is to treat row one as the header, which yields a frame whose columns are named `Unnamed: 0` onward and whose first data row contains the real headers. No exception is raised. A user who does not inspect the frame visually proceeds with a dataset that is off by one row and has lost its column names.

Second, **the two files are organised in opposite directions**. The highway file is long on pollutant and wide on test type: one row per family and pollutant, with separate columns for each test type's result. The nonroad file is wide on pollutant beneath a two-level header, with family attributes distributed on *both* sides of the measurement block. The two require different, and in places opposite, reshaping operations.

Third, **the two encode pollutants and units differently**. The same pollutant appears under different spellings; one panel's results are in grams per brake horsepower-hour and the other's in grams per kilowatt-hour; opacity is a percentage and not a rate at all. Any cross-cutting question requires a reconciliation that the publisher does not provide and that no user has an obligation to perform correctly.

1.3 Why the failure mode matters

The consequence is that this work is duplicated by every user, documented by none, and its errors are invisible.

That last property is the argument for this dataset. A malformed file that refuses to open costs an afternoon. A malformed file that opens cleanly and yields a frame with plausible column names, plausible row counts and wrong contents costs a published result. Every defect described in Section 1.2 has the second shape: a header-row error produces a valid frame, a wide-to-long reshape on a non-unique key produces a larger and still plausible table, and mixing two unit systems in one column produces numbers that differ by a factor of 1.34 rather than by an order of magnitude, which is well inside the range a reader would accept as a real effect.

Section 2.6 documents an instance of exactly this, found during development of CIDEX itself. We include it rather than quietly fixing it because a methods section that claims care is weaker evidence than one that shows what care caught.

1.4 Contributions

1. A harmonized, analysis-ready panel covering both engine categories under one schema, with a machine-readable codebook and Parquet mirrors.
2. A resolved **certification lineage**, linking families across model years — present in the source as unresolved references, not previously published in usable form — together with its depth distribution and an account of the three traversal cases the source requires.
3. A characterisation of two defects in the published record: physically impossible values, and structural discontinuities at archive boundaries that masquerade as trends. Both are flagged in the data rather than described only in prose.
4. A reproducible pipeline with hash-level provenance and a verification procedure that blocks publication until an author has checked the output against the source by hand, with the

checked records pinned by name in the attestation.

5. A worked account of one harmonization defect found and fixed during development, offered as a methods contribution rather than an apology.

1.5 What CIDEX is not

It is not a new measurement. Every value in it appears in EPA’s published files; the contribution is reorganisation, resolution and validation, not observation.

It is not a complete history. EPA holds older model years in separate archive files that this version does not ingest, so the panel’s early years are bounded by what the current file happens to contain rather than by when certification began. Section 5.2 quantifies the boundary and flags every affected row.

It is not a record of in-service emissions. Certification results are laboratory values obtained on a certification fuel and a specified duty cycle. The relationship between them and what an engine emits in service is an open question, and one that a companion project in the same research program takes up from the fuel-specification side; CIDEX supplies the population that question is asked about, and answers none of it by itself.

2 Methods

2.1 Design principles

Four rules govern every decision in the pipeline, and stating them first makes the individual choices predictable rather than arbitrary.

Preserve the source. CIDEX reorganises; it does not correct. Where the published record contains something implausible, the value is retained and flagged. A reader who wants it removed can filter; a reader given a cleaned table cannot recover what was cleaned.

Assert nothing that can be tested. Wherever the pipeline would otherwise rely on domain knowledge — that a column is in particular units, that a key is unique, that a reshape preserved the row count — the belief is written as a check that runs on every build. Section 4.2 gives the clearest case: a units claim that began as an inference and became a test.

Fail loudly. An unmapped pollutant raises rather than being dropped. An attribute that varies within a family raises rather than being resolved by taking the first row. A merge that changes the row count raises rather than proceeding. Silence is the failure mode this dataset exists to remove, so the pipeline is written to be noisy.

Make judgements visible. Every decision that could reasonably have gone the other way — keeping units divergent, retaining NMHCE as distinct from NMHC, treating a self-reference as a terminus — is documented with its reasoning, so a user who disagrees can find the decision rather than discovering it.

2.2 Sources and provenance

Two EPA workbooks, both US Government works without copyright restriction: heavy-duty highway gasoline and diesel engines (model years 2015–2026, 7,804 source rows) and nonroad compression-ignition engines (2011–2027, 10,214 rows).

EPA replaces these files in place each quarter with no changelog and no versioned archive. A dataset built from them is therefore identified by the bytes it was built from, not by the URL. We record the SHA-256 digest of every source file at fetch time along with the retrieval timestamp and byte count; the deposited version carries fixed digests that a rebuilder can verify. If EPA revises a file, a rebuild produces a different digest and the difference surfaces immediately rather than being absorbed into the results.

This is a stronger requirement than it may appear. Without it, “the EPA certification data” names a moving target, two papers citing the same URL can rest on different numbers, and neither can tell. The digest makes the dataset a fixed object that a disagreement can be resolved against.

2.3 Schema and the family key

The target is a single long form keyed on (**panel, model year, engine family, pollutant, test type**), with three supporting tables.

We establish that the engine family name is a **true key** rather than assuming it. In both panels the count of distinct family names equals exactly the count of distinct (model year, family) pairs — 702 and 7,925 respectively — because EPA encodes the model year in the name’s first character. No surrogate key is introduced, so every row remains traceable to EPA’s own record by a value a reader can look up in EPA’s public certificate tool. The equality is re-checked on every build, because it is a property of EPA’s naming convention and not a guarantee.

We separate **family-level** from **configuration-level** attributes, and the need for the separation is measured rather than assumed. In the nonroad source, 1,487 of 7,925 families occupy more than one row — up to 34 rows for a single family — differing by engine code, model, displacement, fuel or test procedure. Retaining these at family level would require an arbitrary row to win.

Table 1 gives the number of families for which each attribute takes more than one value in the source, before the split. These are the columns that would have been silently collapsed by a naive de-duplication.

Table 1. Nonroad attributes conflicting within a family, before the family/configuration split.

Attribute	Families with conflicting values
<code>engine_code</code>	609
<code>test_procedure</code>	428
<code>engine_operation</code>	422
<code>engine_model</code>	278
<code>displacement_l</code>	88
<code>test_procedure_type</code>	80
<code>certification_fuel</code>	35

After the split, the family table’s attribute conflicts number 0, and the highway table’s number 0. The separation is enforced by assertion: the build fails if any attribute remaining in the family table varies within a family.

2.4 Harmonization: the highway panel

The file is read with the header on row two. The pollutant name maps through a controlled vocabulary of 10 terms; an unmapped value raises rather than being dropped, since a silently

discarded pollutant is the precise failure this work exists to prevent. Appendix B lists the vocabulary in full.

The test-type block is unpivoted from 15,608 intermediate rows, yielding one record per family, pollutant and test type. 2,184 rows carry no value in any measurement column and are removed as artefacts of the unpivot rather than facts about an engine; the count is reported rather than absorbed, because a dropped-row count that nobody reports is a dropped-row count that nobody can check. The panel resolves to 702 family rows, 875 configuration rows and 13,424 emission records.

2.5 Harmonization: the nonroad panel

The two-level header is resolved without positional assumptions, which would break on any column insertion. The group row is forward-filled, then a column is classified as a measurement **only if its sub-name is one its group actually contains**. This test is load-bearing: the attribute columns trailing the measurement block inherit a forward-filled group label and would otherwise be misclassified as pollutants. The classifier identifies 22 measurement and 24 attribute columns, and reports 0 unmapped attribute columns.

The wide measurement block unpivots to 224,708 intermediate rows, of which 141,657 are empty cells in the wide layout rather than observations; the wide form is sparse because most families are not tested for most pollutants. 81,445 emission records remain.

The panel also carries 1,606 Family Emission Limit values, which are declared limits rather than test results and must be attached to the corresponding measurement rather than treated as one. Section 2.6 describes what went wrong when they were.

2.6 A defect found during development: the FEL merge

We report this in full because it is the class of error CIDEX exists to prevent, and because the way it was found is more informative than the fix.

FEL values arrive in the nonroad source as rows of the same shape as test results. The first implementation attached them to their measurements by merging on (`model_year`, `engine_family`, `pollutant`). That key is not unique. As Section 2.3 records, 1,487 families occupy more than one source row, up to 34, so a family with several source rows matched every FEL row against every measurement row, and the merge multiplied.

Nothing raised. The merge succeeded, the resulting table had valid types and plausible values, and every individual number in it was a number EPA had published. Only the count was wrong — inflated by roughly two thousand rows against an arithmetic expectation derived independently from the source row counts.

It was found by computing what the row count should be before looking at what it was. That is now a permanent control: the merge is performed on the **source row identifier** rather than the family, and the row count before and after is compared, with a failure raising and naming both numbers. The QA report records the check as `fel_merge_row_count_stable = true`.

Three lessons are worth stating explicitly, because they generalise beyond this dataset. A merge is a join and a join on a non-unique key is a multiplication, which is obvious in the abstract and easy to miss in a pipeline of twenty steps. A transformation's row count is a testable prediction and should be predicted before it is observed. And a check that compares an output to an independently

derived expectation catches a whole class of defects that no amount of type checking or null checking will reach.

2.7 Lineage resolution

5,982 nonroad families name a predecessor family. Resolution is a traversal to each family’s root, with three cases handled explicitly.

Self-reference. 30 families name themselves. EPA appears to use this to denote an unchanged re-certification; the convention is undocumented. A naive traversal treats it as a cycle, and our first implementation reported a large number of spurious cycles for this reason. We treat `parent(x) = x` as a terminus.

Orphans. 568 families name a predecessor absent from the published window, almost all pre-2011 families held in EPA’s archive file. 5,414 edges resolve within the window. Truncated lineages are not erroneous, but they carry a consequence a user must know: **a recorded depth of zero does not always denote the first of its line**. Both counts are published so a user can decide whether to exclude truncated lineages from a depth analysis.

Cycles. Detected with a visited set. Count: 0. The graph is a forest.

The result is 2,541 distinct lineages over 7,925 nonroad families, with mean depth 1.382 and maximum depth 14. The deepest example is family TCPXL18.1NYS (MY2026), rooted at CCPXL18.1NYS. Appendix D gives the full depth distribution; 5,384 families sit at depth one or greater.

2.8 Units

Highway results are certified in g/bhp-hr, nonroad in g/kW-hr, nonroad smoke opacity in percent opacity. **No conversion is performed.** Each record carries its unit system explicitly, and 3 unit labels appear in the panel.

We regard this as a substantive design decision rather than an omission. Normalising to a single system would produce a tidier table in which a user aggregating across panels obtains a plausible and wrong result with nothing to signal the error. The conversion factor is 0.7457, so a mixed aggregation is wrong by a third — large enough to matter and small enough to look like a finding. Leaving the units divergent makes the incompatibility visible at the point of use.

A conversion helper is provided and must be called deliberately. The margin helper, which computes headroom against a standard, raises rather than computing across unit systems: the one operation where a mixed comparison would be most tempting is the one operation that refuses.

2.9 Quality flags

Two flag columns travel with the data rather than with the documentation.

`quality_flag` marks records whose values are physically impossible — in this version, negative certification results. `year_coverage` marks each row’s model year as complete, partially archived or partially forward, so a time series can be filtered without the user having to know where the archive boundaries fall.

The principle in both cases is that a hazard described only in a limitations section is a hazard that will be encountered by users who did not read it. A flag column is read by the code.

3 Data records

3.1 Files

Table 2. Published tables.

Table	Rows	Columns	Contents
<code>cidex_family.csv</code>	8,627	20	One row per (panel, model year, family)
<code>cidex_config.csv</code>	9,794	14	Configurations within a family
<code>cidex_emissions.csv</code>	94,869	17	One row per family, pollutant, test type
<code>cidex_carryover.csv</code>	7,925	6	Resolved lineage: parent, root, depth

Parquet mirrors accompany each table. A machine-readable codebook documents every column; Appendix A lists the schema as published.

The panel covers 108 manufacturers (27 highway, 86 nonroad), 14 pollutants and 3 test types (smoke, steady state, transient).

3.2 Pollutant coverage

Coverage is strongly uneven, and the unevenness is a property of the certification programme rather than of this dataset. Criteria pollutants are measured for nearly every family; greenhouse gases and air toxics are measured for a subset determined by service class and model year.

Table 3. Emission records by pollutant.

Pollutant	Records	Share of panel
CO	14,238	15.0%
NOx	14,238	15.0%
NMHC	14,219	15.0%
PM	14,190	15.0%
CO2	13,196	13.9%
CH4	7,456	7.9%
NMHC_NOx	6,712	7.1%
N2O	4,700	5.0%
smoke_accel	1,853	2.0%
smoke_peak	1,785	1.9%
smoke_lug	1,763	1.9%
HCHO	503	0.5%
NMHCE	13	0.0%
NH3	3	0.0%

NMHCE (Non-Methane Hydrocarbon Equivalent, 13 records) is retained as distinct from NMHC (14,219 records) because it is a different regulatory construct, not a different spelling. Folding it in would destroy information a user cannot reconstruct; keeping it separate costs a filter expression. NH3 (3 records) and HCHO (503 records) are retained on the same principle despite being rare: a pollutant with three records is still a pollutant, and a user studying ammonia slip would rather have three rows than none.

3.3 Model year coverage

Table 4 gives family counts by panel and model year with the coverage classification each row carries. Years marked other than complete are discussed in Section 5.2; they are retained, not removed, and flagged so that a filter can exclude them.

Table 4. Families by panel and model year.

Panel	Model year	Families	Share of interior median	Coverage
highway	2015	3	5%	partial archive
highway	2016	29	45%	partial archive
highway	2017	75	117%	complete
highway	2018	61	95%	complete
highway	2019	61	95%	complete
highway	2020	56	88%	complete
highway	2021	64	100%	complete
highway	2022	72	112%	complete
highway	2023	64	100%	complete
highway	2024	76	119%	complete
highway	2025	75	117%	complete
highway	2026	66	103%	complete
nonroad	2011	66	12%	partial archive
nonroad	2012	521	98%	complete
nonroad	2013	401	75%	complete
nonroad	2014	430	80%	complete
nonroad	2015	454	85%	complete
nonroad	2016	485	91%	complete
nonroad	2017	489	92%	complete
nonroad	2018	496	93%	complete
nonroad	2019	534	100%	complete
nonroad	2020	569	107%	complete
nonroad	2021	583	109%	complete
nonroad	2022	597	112%	complete
nonroad	2023	607	114%	complete
nonroad	2024	597	112%	complete
nonroad	2025	558	104%	complete
nonroad	2026	530	99%	complete
nonroad	2027	8	2%	partial forward

Usable ranges, defined as the contiguous span of years at or above 50% of the panel’s interior median, are highway MY2017–2026 and nonroad MY2012–2026.

3.4 Nonroad standard tiers

Table 5. Nonroad families by emission standard tier.

Tier	Families	Share of nonroad panel
Tier 4 (Final or Phase In)	5,626	71.0%
Tier 3	983	12.4%
Interim Tier 4	828	10.4%
Tier 2	488	6.2%

The distribution is a snapshot of the currently published window rather than a history of adoption: the window is bounded below by the archive boundary described in Section 5.2, so earlier tiers are under-represented by construction and the table should not be read as a technology transition curve.

4 Technical validation

4.1 Automated checks

Ten checks execute on every build. All pass.

Table 6. Integrity checks.

Check	Result
<code>family_key_unique</code>	pass
<code>no_emission_row_without_family</code>	pass
<code>units_never_null</code>	pass
<code>no_mixed_units_within_panel</code>	pass
<code>highway_units_evidenced_by_standards</code>	pass
<code>highway_units_labelled</code>	pass
<code>nonroad_units_correct</code>	pass
<code>carryover_no_true_cycles</code>	pass
<code>every_negative_value_is_flagged</code>	pass
<code>flagged_share_below_0pt1pct</code>	pass

These establish **internal consistency**: that the key is unique, that no emission record refers to a family that does not exist, that units are labelled and consistent within a panel, that the lineage graph is acyclic, and that every anomalous value is flagged. They cannot establish correspondence with EPA’s record, which is what Section 4.3 is for.

Two of the checks deserve comment because they encode decisions rather than truisms. `every_negative_value_is_flagged` asserts that the pipeline has not silently dropped the anomalies it found. `flagged_share_below_0pt1pct` is a tripwire: it does not assert that the flagged share is correct, but it fails if the share ever jumps, which would indicate that a change to the pipeline had begun flagging a class of values it previously accepted.

4.2 Units established by evidence, not assertion

The highway source states its units nowhere; its headers are bare. That highway results are in g/bhp-hr was initially an inference from domain knowledge, which is to say an assumption with a good pedigree and no test behind it.

We replaced it with a test. United States heavy-duty highway standards have known values in g/bhp-hr, and the modal certification standard per pollutant in the panel is compared against them.

Table 7. Units evidence: known heavy-duty highway standards against the modal value in CIDEX.

Pollutant	Known standard (g/bhp-hr)	Modal value in CIDEX	Same standard in g/kW-hr
NOx	0.2	0.2	0.268
PM	0.01	0.01	0.013
NMHC	0.14	0.14	0.188
CO	15.5	15.5	20.786

The final column is why the test cannot pass by coincidence: expressed in the other unit system the same standards are an order of magnitude distant in every case, so a panel mislabelled as g/bhp-hr would fail the comparison rather than pass it weakly. The check runs on every build, so a future source change that altered the unit basis would break the build rather than quietly change the meaning of a column.

4.3 Manual verification

No automated check can establish correspondence with the source. A pipeline can verify that it did what it was told; only a person can verify that what it was told was right.

The pipeline ships five named spot-checks, each probing a distinct failure mode, which an author looks up in EPA’s public certificate tool and compares by hand.

Table 8. Pinned spot-checks.

Family	MY	Panel	Pollutant	Test	CIDEX value	Failure mode probed
MSZXH05.23FD	2021	highway	NOx	transient	0.17 g/bhp-hr	highway, transient NOx — the standard case
FNGCH0466AEA	2015	highway	PM	steady state	0.01 g/bhp-hr	highway, earliest model year present — tests the MY2015-16 archive boundary
LCEXL60.0AAB	2020	nonroad	NOx	steady state	0.48 g/kW-hr	nonroad, steady-state NOx — tests the wide-to-long unpivot
TCPXL32.1NZS	2026	nonroad	NMHC	steady state	0.08 g/kW-hr	deepest carryover lineage (depth 14, root CCPXL32.0NZS) — tests lineage resolution
CDICL05.8HTA	2012	nonroad	NOx	steady state	-1.0 g/kW-hr	a flagged negative value — is this real in EPA’s own record?

Publication is blocked until this is signed. The attestation **pins which records were checked**, by family name, because the selector originally chose a median row and its output moves when the row set changes; without pinning, a signature given against one set of records could come to refer to records nobody had examined. The gate compares the pinned names against the current selection and fails on drift.

4.4 What the validation does not establish

It does not establish that EPA’s published values are correct. CIDEX inherits whatever is in the source, and Section 5.1 documents values that are certainly wrong in the source and are preserved anyway.

It does not establish that the five checked records generalise. They were chosen to span distinct failure modes rather than to be a random sample, which is the right design for catching structural errors and the wrong design for estimating an error rate. We make no error-rate claim.

It does not establish completeness against EPA’s full certification history, for the archive reason in Section 1.5.

5 Anomalies in the published record

5.1 Physically impossible values

50 records (0.0527% of the panel, 18 families, all nonroad) carry a **negative** certification result. An emission rate cannot be negative.

We publish them unaltered, flagged. Correcting them would break the claim that CIDEX faithfully reorganises a published record, and would conceal a finding: either the source contains data-entry errors worth raising with EPA, or an encoding convention exists — a sentinel for “not applicable”, perhaps — that neither party has documented. Both possibilities are more useful to a reader than a quietly cleaned column, and a reader who wants them gone has one filter expression to write.

The share is small enough that it would not disturb an aggregate and large enough that it would disturb a minimum. That asymmetry is the argument for flagging rather than documenting: a user computing a minimum or a lower percentile needs to know, and a user computing a mean does not, and neither can be relied upon to have read a limitations section.

One flagged record is included among the pinned spot-checks in Table 8, so the verification procedure confirms that the negative value is present in EPA’s own record rather than introduced by the pipeline.

5.2 Archive boundaries that resemble trends

Both panels thin sharply at their edges, for two unrelated reasons. EPA holds older model years in **separate archive files**, so the earliest year of each panel contains only whichever families happen to sit in the current file. Separately, the newest model year is still being certified when the file is retrieved, so its count is partial for a reason that has nothing to do with the fleet.

Table 9. Model years with incomplete coverage.

Panel	Model year	Families	Share of interior median	Coverage
highway	2015	3	5%	partial archive
highway	2016	29	45%	partial archive
nonroad	2011	66	12%	partial archive
nonroad	2027	8	2%	partial forward

A time series spanning these years reads a filing artefact as a collapse or a boom. The highway panel’s MY2015 count is 5% of the interior median; a chart drawn without the flag would show a catastrophic fall in heavy-duty certification followed by an equally implausible recovery, and nothing in the data would contradict it.

We flag every affected row rather than documenting the hazard in prose alone, and report usable ranges: highway MY2017–2026, nonroad MY2012–2026.

5.3 Depth zero is ambiguous

The 568 orphan edges of Section 2.7 produce a subtler hazard. A family whose declared predecessor lies outside the published window resolves to itself as root and records depth zero, which is indistinguishable in the table from a family that genuinely begins its line.

2,541 families sit at depth zero. Some are first-of-line; some are truncated. The carryover table carries the orphan status so the two can be separated, and a depth analysis that does not separate them will understate lineage length by an amount that grows with how much of the history sits in the archive.

5.4 An undocumented convention

The 30 self-referencing families of Section 2.7 are worth recording as a finding in their own right. A family naming itself as its own predecessor is not a data error: it appears to be a deliberate encoding for an unchanged re-certification. But it is documented nowhere, and the natural implementation of a lineage traversal treats it as a cycle and either loops forever or reports a corrupt graph.

We report it because the next person to resolve this structure will otherwise rediscover it by debugging, and because it is a small example of a general point about open government data: publication of the bytes is not publication of the semantics, and the gap between them is where reuse costs accumulate.

6 Usage notes

6.1 Three cautions

Do not compare across panels without converting units. Filter on `panel`, or convert deliberately using the provided helper. The conversion factor is 0.7457; an unconverted cross-panel aggregate is wrong by about a third.

Exclude incomplete years from time series. Filter `year_coverage == "complete"`, or restrict to the usable ranges given in Section 3.3.

Decide about flagged values explicitly. Filter on `quality_flag` if physically plausible values are required. Do not assume the default is what you want; the default is the source.

6.2 What the panel supports

The harmonized panel supports the questions the source makes expensive: the distribution of certification results within and across manufacturers, the relationship between certified result and applicable standard, the composition of the certified population by tier and displacement, and cross-panel comparisons where the unit discipline is observed.

The lineage table supports questions the panel alone cannot answer. How long does a certification family persist? How do technology changes propagate through successive re-certifications? Does lineage depth differ systematically across power categories, tiers or manufacturers? Is a deep lineage associated with a certification result that changes slowly? These are questions about the regulatory process rather than about engines, and the resolved lineage is what makes them askable from public data.

Because the family name is EPA's own and no surrogate key is introduced, any row can be taken back to EPA's certificate tool and checked. A result computed from CIDEX is falsifiable against the source by a reader who has never run the pipeline.

6.3 Limitations

EPA's archive files are not ingested in this version, which bounds the panel's history and produces the depth-zero ambiguity of Section 5.3.

Manufacturer names are not normalised across years. The same corporate entity may appear under more than one spelling, so a manufacturer-level aggregate requires the user to decide on a normalisation. We do not impose one, because the right normalisation depends on whether the question is about corporate groups, legal entities or brands, and that is the user's question to answer.

Nonroad records carry no pollutant-specific standard, because the source states the standard only at family level. Inferring a per-pollutant standard from the family-level value would be a guess presented as data.

Certification results are laboratory values on a certification fuel and a specified duty cycle. They are not in-service emissions, and the difference is not a small correction.

The panel is a snapshot of a file EPA replaces quarterly. It is identified by digest and should be cited by version.

7 Reuse and extension

Three extensions are straightforward and would each add substantially to the panel's value.

Ingest the archive files. This would extend both panels backward, close the depth-zero ambiguity for most lineages, and turn the tier distribution in Table 5 into something that could legitimately be read as an adoption curve.

Normalise manufacturer identity. A curated crosswalk, published separately so that users can disagree with it, would make manufacturer-level analysis possible without each user building their own.

Link to other public records. The engine family is a key that appears in other federal datasets. Joining on it would connect certification to recall, in-use testing and enforcement records, and the join is cheap precisely because CIDEX keeps EPA's own key rather than inventing one.

We publish the pipeline under a permissive licence to make these extensions possible without permission or coordination.

Data availability

The dataset is deposited on Zenodo under the concept DOI 10.5281/zenodo.22761791, which always resolves to the newest version; the version described here is 10.5281/zenodo.22833302. The dataset is released under CC BY 4.0. All processed tables are in the repository under `data/processed/`, with Parquet mirrors and a machine-readable codebook. The source files are US Government works published by EPA and are identified in the provenance log by URL, retrieval date, byte count and SHA-256 digest.

Code availability

The complete pipeline is at <https://github.com/osariemenimafidon/cidex> under the MIT licence, with a technical report, codebook, verification checklist and provenance log. A single command rebuilds every published figure from the source files and diffs the result against the deposited values.

Competing interests

The author declares no competing interests.

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Use of AI assistance

Pipeline code, table generation and manuscript drafting were produced with AI assistance. The harmonization decisions, the verification spot-checks against EPA’s certificate tool and the interpretation of the record’s structure were made and checked by the author, who is responsible for the content.

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Appendix A: published schema

Column names as they appear in the published files.

`cidex_family.csv` (20 columns)

`panel, manufacturer, model_year, engine_family, certificate_no, date_issued, engine_cycle, fuel_type, fuel_metering_system, introduction_date, useful_life, intended_service_class, carryover_family, power_category, regulation, tier, compliance_standard, non_aftertreatment_device, aftertreatment_device, year_coverage`

`cidex_config.csv` (14 columns)

`panel, model_year, engine_family, engine_code, engine_test_model, engine_id, test_fuel, test_date, engine_model, displacement_l, certification_fuel, engine_operation, test_procedure, test_procedure_type`

`cidex_emissions.csv` (17 columns)

`panel, model_year, engine_family, engine_code, engine_test_model, pollutant, test_type, cert_result, adj_result, standard, fel, fcl, df_type, df_value, units, quality_flag, year_coverage`

`cidex_carryover.csv` (6 columns)

`panel, model_year, engine_family, carryover_family, lineage_root, lineage_depth`

Appendix B: pollutant vocabulary

Source spellings and their canonical forms. An unmapped source value raises rather than being dropped.

Ammonia → NH₃, Carbon Dioxide → CO₂, Carbon Monoxide → CO, Formaldehyde → HCHO, Methane → CH₄, Nitrogen Oxides → NO_x, Nitrous Oxide → N₂O, Non-Methane Hydrocarbon Equivalent → NMHCE, Non-Methane Hydrocarbons → NMHC, Particulate Matter → PM

Appendix C: test types and units

Test types: `smoke`, `steady_state`, `transient`.

Unit labels present: `g/bhp-hr`, `g/kW-hr`, `pct opacity`.

Conversion factor between the two rate systems: 0.7456998716 kW per bhp. The pipeline's margin helper raises rather than computing a headroom across unit systems.

Appendix D: lineage depth distribution

Depth	Families	Share of nonroad panel
0	2,541	32.1%
1	3,719	46.9%

Depth	Families	Share of nonroad panel
2	583	7.4%
3	295	3.7%
4	211	2.7%
5	157	2.0%
6	114	1.4%
7	67	0.8%
8	58	0.7%
9	50	0.6%
10	41	0.5%
11	36	0.5%
12	23	0.3%
13	16	0.2%
14	14	0.2%

Depth zero includes both first-of-line families and lineages truncated at the archive boundary; see Section 5.3.

Preprint generated 2026-09-15. Every quantity is interpolated from the pipeline’s own statistics and quality-assurance files; none is transcribed by hand.